



2015
PRELIMINARY
EXAMINATION

Student Number: _____

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 6–9.

Total Marks – 65

Section I

Page 2

5 marks

- Attempt Questions 1–5
- Allow about 10 minutes for this section

Section II

Pages 3–6

60 marks

- Attempt Questions 6–9
- Allow about 1 hours 50 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1 What is the value of $\sqrt{\frac{9.5+3.7}{9.5-3.7}}$, correct to 3 significant figures?

- (A) 1.50 (B) 1.508 (C) 1.509 (D) 1.51
-

2 Which of the following is equivalent to $x^2 - 5x - 6$?

- (A) $(x+2)(x-3)$ (B) $(x-1)(x+6)$ (C) $(x+1)(x-6)$ (D) $(x-2)(x+3)$
-

3 What are the coordinates of the focus of the parabola $x^2 = 4(y+1)$?

- (A) (0,−1) (B) (0,0) (C) (0,1) (D) (0,4)
-

4 Which of the following does **not** represent a function?

- (A) $x = 1$ (B) $y = 1$ (C) $x = \sqrt{y}$ (D) $y = \sqrt{x}$
-

5 The lines $2x - ky = 0$ and $x - 3y = 0$ are perpendicular. What is the value of k ?

- (A) $-\frac{1}{6}$ (B) $-\frac{2}{3}$ (C) $-\frac{3}{2}$ (D) -6
-

Section II

60 marks

Attempt Questions 6–9

Allow about 1 hours 50 minutes for this section

Start each question on a new page. Extra writing pages are available.

All necessary working should be shown in every question.

Question 6

(15 marks)

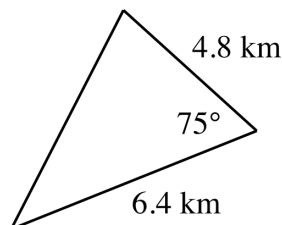
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- (a) (i) Find the equation of the axis of symmetry of the parabola $y = x^2 - 4x - 5$. **1**
- (ii) Find the minimum value of the expression $x^2 - 4x - 5$. **1**

- (b) The function $f(x)$ is given by: **2**
- $$f(x) = \begin{cases} x^2 - 5 & \text{for } x \leq 2 \\ 7 - 2x & \text{for } x > 2 \end{cases}$$

Find $f(-5) + f(3)$.

(c)



A triangular park has two sides of length 6.4 km and 4.8 km. The angle between these sides is 75° .

- (i) Find the length of the third side of the park. Give your answer in kilometres, correct to 1 decimal place. **2**
- (ii) Find the area of the park. Give your answer correct to the nearest square kilometre. **1**
- (d) Solve the following inequations.
- (i) $2x^2 - x - 28 < 0$ **2**
- (ii) $|2x - 1| > 3$ **2**

Question 6 (continued)

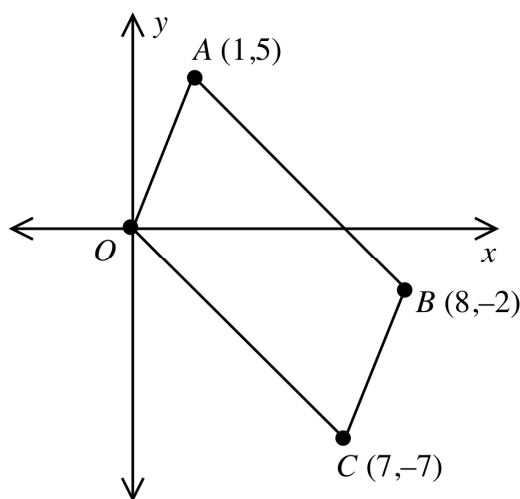
- (e) (i) Draw a large, neat sketch of the function $y = \sqrt{4 - x^2}$. Indicate where the curve crosses the coordinate axes. 2
- (ii) Write down the domain and range of the function $y = \sqrt{4 - x^2}$. 2

Question 7

(15 marks)

Start a new page

- (a) In the diagram below, $O(0,0)$, $A(1,5)$, $B(8,-2)$ and $C(7,-7)$ are the vertices of the quadrilateral $OABC$.



- (i) Find the midpoint of the interval joining AC . 1
- (ii) Find the gradient of AB . 1
- (iii) Show that the equation of AB is $x + y - 6 = 0$. 1
- (iv) Find the exact length of AB . 2
- (v) Show that AB is parallel to OC . 1
- (vi) What type of quadrilateral is $OABC$? Use appropriate reasoning and calculations to justify your answer. 2
- (vii) Find the perpendicular distance from O to AB . Give your answer in exact form. 2
- (viii) Hence find the area of quadrilateral $OABC$. Give your answer in exact form. 2
- (b) Find the values of A , B and C , given that $A(x^2 + 1)^2 + B(x^2 + 1) + C \equiv 2x^4 + 3x^2 - 3$. 3

Question 8

(15 marks)

[Start a new page](#)

- (a) $A(-2,5)$ and $B(1,1)$ are two points. Find the coordinates of the point $P(x,y)$ such that B is the midpoint of AP . **2**
- (b) (i) Show that $(p + q)^2 - 4pq = (p - q)^2$. **1**
- (ii) Prove that the roots of the equation $px^2 - (p + q)x + q = 0$ are rational. **2**
- The quadratic equation $2x^2 + 4x - 1 = 0$ has roots α and β . Find the values of:
- (iii) $\alpha + \beta$ **1**
- (iv) $\alpha\beta$ **1**
- (v) $|\alpha - \beta|$ **2**
- (c) Consider the function $f(x) = 2^x + 2^{-x}$.
- (i) Show algebraically that $f(x)$ is an even function. **2**
- (ii) Sketch the graph of $y = f(x)$. **1**
- (d) Show that the line $3x + 4y - 9 = 0$ is a tangent to the circle $(x - 1)^2 + (y + 1)^2 = 4$. **3**

Question 9

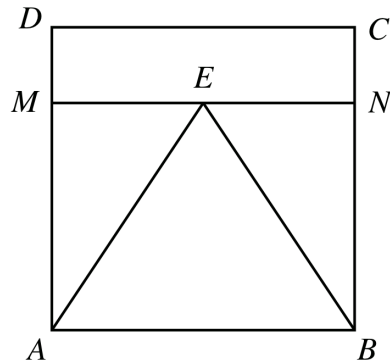
(15 marks)

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- (a) Solve the equation $\sqrt{3} \tan \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. **2**
- (b) Find the value of m in the quadratic equation $2x^2 - 27x + m = 0$ if one root is double the other. **2**
- (c) Prove that $\sin^2 \alpha \cos^2 \beta - \cos^2 \alpha \sin^2 \beta = \sin^2 \alpha - \sin^2 \beta$. **3**

Question 9 (continued)

(d)



In the diagram above, $ABCD$ is a square of side 2 cm. E is the point inside the square such that $\triangle ABE$ is an equilateral triangle. The line MN through E is perpendicular to both AD and BC .

(i) Copy the diagram onto your answer sheet.

(ii) Show that $\angle ECN = 75^\circ$. **3**

(iii) Show that $\tan 75^\circ = 2 + \sqrt{3}$. **2**

(e) Solve the equation $(x^2 - x - 10)(x^2 - x - 16) = 40$. **3**

End of paper

2 UNIT MATHEMATICS
2015 PRELIMINARY EXAMINATION

SECTION I

$$1 \quad \sqrt{\frac{9.5+3.7}{9.5-3.7}} = \sqrt{\frac{13.2}{5.8}}$$

$$= 1.50859606$$

$$\approx 1.51$$

1. **D**

$$2 \quad x^2 - 5x - 6 = (x + 1)(x - 6)$$

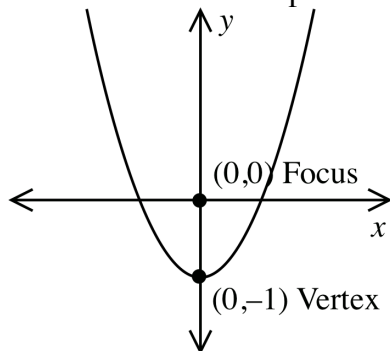
2. **C**

$$3 \quad x^2 = 4(y + 1)$$

$$x^2 = 4.1.(y + 1)$$

$\therefore$ Focal length = 1 unit
 Parabola is concave up.

3. **B**



Vertex $\equiv (0, -1)$

$\therefore$ Focus $\equiv (0, 0)$

- 4 The graph of $x = 1$ represents a vertical line.
 For this value of x there is more than one corresponding value of y .
 $\therefore x = 1$ is not a function.

4. **A**

5

5. **B**

$$\text{Gradient of line 1} = \frac{-a}{b}$$

$$= \frac{-2}{-k}$$

$$= \frac{2}{k}$$

$$\text{Gradient of line 2} = \frac{-a}{b}$$

$$= \frac{-1}{-3}$$

$$= \frac{1}{3}$$

Since the lines are perpendicular we have:

$$m_1 \times m_2 = -1$$

$$\frac{2}{k} \times \frac{1}{3} = -1$$

$$\frac{2}{3k} = -1$$

$$2 = -3k$$

$$k = -\frac{2}{3}$$

SECTION II

QUESTION 6

- (a) (i) Equation of axis of symmetry is:

$$\begin{aligned}x &= \frac{-b}{2a} \\&= \frac{-(-4)}{2(1)} \\&= 2\end{aligned}$$

- (ii) When $x = 2$:

$$\begin{aligned}y &= (2)^2 - 4(2) - 5 \\&= 4 - 8 - 5 \\&= -9\end{aligned}$$

$\therefore$ Minimum value $= -9$

(b) $f(-5) + f(3) = [(-5)^2 - 5] + [7 - 2(3)]$
 $= [25 - 5] + [7 - 6]$
 $= 20 + 1$
 $= 21$

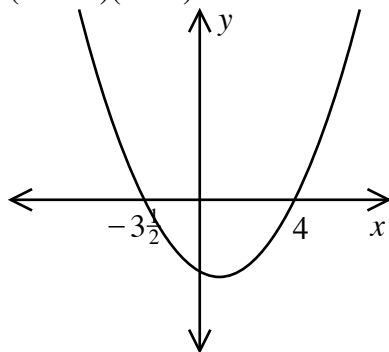
- (c) (i) Let d be the length of the third side.

$$\begin{aligned}d^2 &= 4.8^2 + 6.4^2 - 2 \times 4.8 \times 6.4 \times \cos 75^\circ \\&= 48.09815737 \\d &= 6.935283546\end{aligned}$$

$\therefore$ The length of the third side is 6.9 km.

(ii) Area $= \frac{1}{2} \times 4.8 \times 6.4 \times \sin 75^\circ$
 $= 14.83662069$
 $= 15 \text{ km}^2$

(d) (i) $2x^2 - x - 28 < 0$
 $(2x + 7)(x - 4) < 0$



$$\therefore -3\frac{1}{2} < x < 4$$

(ii) When $2x - 1 \geq 0$ (or $x \geq \frac{1}{2}$):

$$2x - 1 > 3$$

$$2x > 4$$

$$x > 2$$

Since $x \geq \frac{1}{2}$, the solution $x > 2$ is valid.

When $2x - 1 < 0$ (or $x < \frac{1}{2}$):

$$-(2x - 1) > 3$$

$$-2x + 1 > 3$$

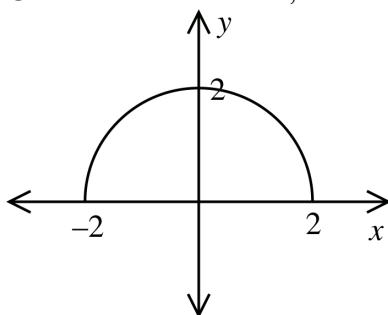
$$-2x > 2$$

$$x < -1$$

Since $x < \frac{1}{2}$, the solution $x < -1$ is valid.

$\therefore$ The solution is $x < -1$ and $x > 2$.

(e) (i) Curve is a semi-circle, on the positive side of the x -axis.



(ii) Domain: $\{x: -2 \leq x \leq 2\}$

Range: $\{y: 0 \leq y \leq 2\}$

QUESTION 7

(a) (i) Let M be the midpoint of the interval AC .

$$\begin{aligned} M &\equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &\equiv \left(\frac{1 + 7}{2}, \frac{5 - 7}{2} \right) \\ &\equiv (4, -1) \end{aligned}$$

(ii) Gradient of $AB = \frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{-2 - 5}{8 - 1}$$

$$= \frac{-7}{7}$$

$$= -1$$

(iii) Equation of AB :

$$\begin{aligned}\frac{y-y_1}{x-x_1} &= \frac{y_2-y_1}{x_2-x_1} \\ \frac{y-5}{x-1} &= \frac{-2-5}{8-1} \\ \frac{y-5}{x-1} &= \frac{-7}{7} \\ \frac{y-5}{x-1} &= -1 \\ y-5 &= -1(x-1) \\ y-5 &= -x+1 \\ x+y-6 &= 0\end{aligned}$$

(iv)

$$\begin{aligned}AB^2 &= (x_1-x_2)^2 + (y_1-y_2)^2 \\ &= (1-8)^2 + (5+2)^2 \\ &= 7^2 + 7^2 \\ &= 98 \\ AB &= \sqrt{98} \\ &= 7\sqrt{2} \text{ units}\end{aligned}$$

(v) Gradient of $AB = -1$

$$\begin{aligned}\text{Gradient of } OC &= \frac{y_2-y_1}{x_2-x_1} \\ &= \frac{0+7}{0-7} \\ &= \frac{7}{-7} \\ &= -1\end{aligned}$$

$\therefore AB$ is parallel to OC

(vi) Let N be the midpoint of the interval OB .

$$\begin{aligned}N &\equiv \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right) \\ &\equiv \left(\frac{0+8}{2}, \frac{0-2}{2} \right) \\ &\equiv (4, -1)\end{aligned}$$

Since M and N are the same point, the diagonals bisect each other.

$\therefore OABC$ is a parallelogram.

(vii)

$$\begin{aligned}\text{Distance} &= \left| \frac{ah+bk+c}{\sqrt{a^2+b^2}} \right| \\ &= \left| \frac{1.0+1.0-6}{\sqrt{1^2+1^2}} \right| \\ &= \left| \frac{-6}{\sqrt{2}} \right| \\ &= \frac{6}{\sqrt{2}} \text{ units}\end{aligned}$$

(viii) Since $OABC$ is a parallelogram,

$$\text{Area} = bh$$

$$= 7\sqrt{2} \times \frac{6}{\sqrt{2}}$$

$$= 42 \text{ square units}$$

$$\begin{aligned} \text{(b)} \quad 2x^4 + 3x^2 - 3 &\equiv A(x^2 + 1)^2 + B(x^2 + 1) + C \\ &\equiv A(x^4 + 2x^2 + 1) + B(x^2 + 1) + C \\ &\equiv Ax^4 + 2Ax^2 + A + Bx^2 + B + C \\ &\equiv Ax^4 + (2A + B)x^2 + (A + B + C) \end{aligned}$$

$$A = 2$$

$$2A + B = 3$$

$$2(2) + B = 3$$

$$4 + B = 3$$

$$B = -1$$

$$A + B + C = -3$$

$$2 - 1 + C = -3$$

$$1 + C = -3$$

$$C = -4$$

$$\therefore A = 2, B = -1 \text{ and } C = -4$$

QUESTION 8

(a) We have:

$$\frac{-2+x}{2} = 1$$

$$-2+x = 2$$

$$x = 4$$

and:

$$\frac{5+y}{2} = 1$$

$$5+y = 2$$

$$y = -3$$

$$\therefore P \equiv (4, -3)$$

$$\begin{aligned} \text{(b) (i)} \quad (p+q)^2 - 4pq &= p^2 + 2pq + q^2 - 4pq \\ &= p^2 - 2pq + q^2 \\ &= (p-q)^2 \end{aligned}$$

(ii) For rational roots, we need the discriminant to be a perfect square.

$$\Delta = b^2 - 4ac$$

$$= [-(p+q)]^2 - 4 \times p \times q$$

$$= (p+q)^2 - 4pq$$

$$= (p-q)^2$$

$\therefore$ The roots of the equation are rational.

(iii)

$$\begin{aligned}\alpha + \beta &= \frac{-b}{a} \\ &= \frac{-4}{2} \\ &= -2\end{aligned}$$

(iv)

$$\begin{aligned}\alpha\beta &= \frac{c}{a} \\ &= \frac{-1}{2} \\ &= -\frac{1}{2}\end{aligned}$$

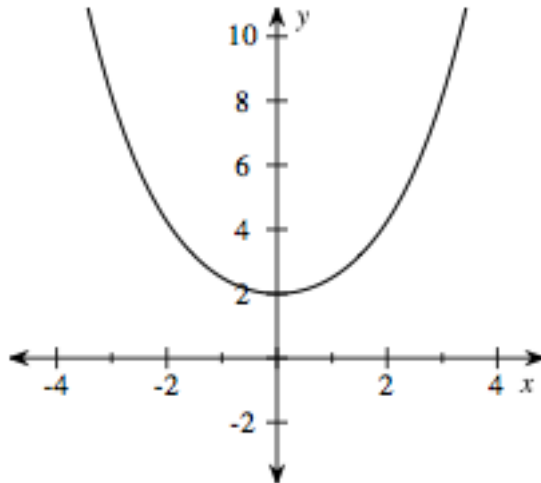
(v)

$$\begin{aligned}(\alpha - \beta)^2 &= (\alpha + \beta)^2 - 4\alpha\beta \\ &= (-2)^2 - 4(-\frac{1}{2}) \\ &= 4 + 2 \\ &= 6\end{aligned}$$

$$\begin{aligned}|\alpha - \beta| &= \left| \sqrt{(\alpha - \beta)^2} \right| \\ &= \left| \sqrt{6} \right| \\ &= \sqrt{6}\end{aligned}$$

- (c) (i) $f(x) = 2^x + 2^{-x}$
 $f(-x) = 2^{(-x)} + 2^{-(-x)}$
 $= 2^{-x} + 2^x$
 $= f(x)$
 $\therefore f(x)$ is an even function

(ii)



(d) We have:

$$(x-1)^2 + (y+1)^2 = 4 \dots [1]$$

and:

$$3x + 4y - 9 = 0$$

$$3x = 9 - 4y$$

$$x = 3 - \frac{4}{3}y \dots [2]$$

Substituting [2] into [1] gives:

$$\left(3 - \frac{4}{3}y - 1\right)^2 + (y+1)^2 = 4$$

$$\left(2 - \frac{4}{3}y\right)^2 + (y+1)^2 = 4$$

$$4 - \frac{16}{3}y + \frac{16}{9}y^2 + y^2 + 2y + 1 = 4$$

$$\frac{25}{9}y^2 - \frac{10}{3}y + 5 = 4$$

$$\frac{25}{9}y^2 - \frac{10}{3}y + 1 = 0$$

$$25y^2 - 30y + 9 = 0$$

$$(5y+3)^2 = 0$$

Since there is only one solution to this equation, the straight line only touches the circle in one point.

$\therefore$ The line is a tangent to the circle.

or

Centre of circle is (1, -1) and radius is 2 units.

Using perpendicular distance formula:

$$\begin{aligned} d &= \left| \frac{ah + bk + c}{\sqrt{a^2 + b^2}} \right| \\ &= \left| \frac{3 \cdot 1 + 4 \cdot (-1) - 9}{\sqrt{3^2 + 4^2}} \right| \\ &= \left| \frac{3 - 4 - 9}{\sqrt{25}} \right| \\ &= \left| \frac{-10}{5} \right| \\ &= 2 \end{aligned}$$

Since this is equal to the radius of the circle, the line $3x + 4y - 9 = 0$ only touches the circle in one point.

$\therefore$ The line is a tangent to the circle.

QUESTION 9

(a)

$$\sqrt{3} \tan \theta + 1 = 0$$

$$\sqrt{3} \tan \theta = -1$$

$$\tan \theta = -\frac{1}{\sqrt{3}}$$

$$\theta = (180 - 30)^\circ \text{ or } (360 - 30)^\circ$$

$$\theta = 150^\circ \text{ or } 330^\circ$$

- (b) Let the roots be α and 2α .

Therefore:

$$\begin{aligned}\alpha + 2\alpha &= \frac{-b}{a} \\ 3\alpha &= \frac{-(-27)}{2} \\ &= \frac{27}{2} \\ \alpha &= \frac{9}{2}\end{aligned}$$

and:

$$\begin{aligned}\alpha \times 2\alpha &= \frac{c}{a} \\ 2\alpha^2 &= \frac{m}{2} \\ m &= 4\alpha^2 \\ &= 4\left(\frac{9}{2}\right)^2 \\ &= 81\end{aligned}$$

$$\begin{aligned}\text{(c) LHS} &= \sin^2\alpha \cos^2\beta - \cos^2\alpha \sin^2\beta \\ &= \sin^2\alpha (1 - \sin^2\beta) - (1 - \sin^2\alpha) \sin^2\beta \\ &= \sin^2\alpha (1 - \sin^2\beta) - \sin^2\beta (1 - \sin^2\alpha) \\ &= \sin^2\alpha - \sin^2\alpha \sin^2\beta - \sin^2\beta + \sin^2\alpha \sin^2\beta \\ &= \sin^2\alpha - \sin^2\beta \\ &= \text{RHS}\end{aligned}$$

$$\begin{aligned}\text{(d) (ii) } AB &= BC = 2 \text{ cm} && \text{(Equal sides of square } ABCD) \\ AB &= EB = 2 \text{ cm} && \text{(Equal sides of equilateral } \triangle ABE) \\ \therefore BC &= EB = 2 \text{ cm} \\ \therefore \triangle EBC &\text{ is isosceles} \\ \therefore \angle CEB &= \angle ECB && \text{(Equal angles opposite equal sides)} \\ \text{Since } \angle EBC &= 30^\circ, \text{ we have:} && \text{(Angle sum of triangle)} \\ \angle ECB + \angle CEB + 30^\circ &= 180^\circ \\ 2\angle ECB + 30^\circ &= 180^\circ \\ 2\angle ECB &= 150^\circ \\ \angle ECB &= 75^\circ\end{aligned}$$

- (iii) Since E is the midpoint of MN we have $ME = EN = 1$ cm.

Using Pythagoras' theorem:

$$\begin{aligned}EB^2 &= EN^2 + NB^2 \\ 2^2 &= 1^2 + NB^2 \\ 4 &= 1 + NB^2 \\ NB^2 &= 3 \\ NB &= \sqrt{3}\end{aligned}$$

Therefore:

$$\begin{aligned}CN &= CB - NB \\ &= 2 - \sqrt{3}\end{aligned}$$

Now:

$$\begin{aligned}\tan \angle ECN &= \frac{EN}{CN} \\ \tan 75^\circ &= \frac{1}{2-\sqrt{3}} \\ &= \frac{1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} \\ &= \frac{2+\sqrt{3}}{4-3} \\ &= 2+\sqrt{3}\end{aligned}$$

(e) $(x^2 - x - 10)(x^2 - x - 16) = 40$

Let $t = x^2 - x$

$$(t-10)(t-16) = 40$$

$$t(t-16) - 10(t-16) = 40$$

$$t^2 - 16t - 10t + 160 = 40$$

$$t^2 - 26t + 120 = 0$$

$$(t-20)(t-6) = 0$$

$$t = 6 \text{ or } 20$$

Therefore:

$$x^2 - x = 6$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3 \text{ or } -2$$

and:

$$x^2 - x = 20$$

$$x^2 - x - 20 = 0$$

$$(x-5)(x+4) = 0$$

$$x = 5 \text{ or } -4$$

$\therefore$ Solutions are $x = -4, -2, 3$ and 5 .